

CHAPTER 5: STATES OF MATTER (JEE FOCUS)

Topics Covered:

- Intermolecular forces
- Gas laws
- Ideal gas equation
- Kinetic theory of gases
- Real gases
- Liquefaction of gases
- Liquid state

Core Concepts:

- Intermolecular forces → attraction between molecules
- Strength: H-bond > dipole > dispersion
- Boyle's law → $P \propto 1/V$
- Charles law → $V \propto T$
- Avogadro law → $V \propto n$
- Ideal gas → no interactions
- $PV = nRT$
- Kinetic energy $\propto$ temperature
- RMS speed $\propto \sqrt{T/M}$
- Real gases → deviate at high P, low T
- Liquefaction → cooling + compression
- Surface tension → cohesive forces
- Viscosity → resistance to flow

Key Formulas:

- $PV = nRT$
- $P_1V_1 = P_2V_2$
- $V_1/T_1 = V_2/T_2$
- $u_{rms} = \sqrt{(3RT/M)}$
- $Z = PV/nRT$
- $KE = (3/2)RT$

Important Patterns:

- High T, low P → ideal behavior
- Heavy gas → slower speed
- Higher IMF → higher boiling point
- H-bond → strongest (N, O, F)
- $Z \neq 1$ → real gas
- Compression → deviation ↑
- Liquids → high density, low compressibility

Common Mistakes:

- Wrong units in gas equation
- Forgetting Kelvin temperature
- Confusing IMF strength order
- Using ideal equation for real gas blindly
- Ignoring molar mass in speed

- Wrong interpretation of Z

Examples (Important Cases):

- Compare speeds of gases
- Gas law numericals
- Boiling point order (IMF)
- Deviation from ideal gas
- Surface tension examples

Exam Tips:

- Convert temperature to Kelvin
- Check units before solving
- Memorize gas laws
- Use shortcuts for ratios
- Practice numericals (very important)
- Focus on IMF questions
- Identify ideal vs real gas quickly